

Homework #1

1 a) Derive the gradient update rule for the weight(s) for a single neuron with one input feature (i.e., field, or attribute) and the sigmoid activation function.

(a) $L = \frac{1}{2}(\hat{y} - y)^2 \quad \frac{dL}{d\hat{y}} = \hat{y} - y$

$\hat{y} = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1} \quad \frac{d}{dz} \hat{y} = \frac{d}{dz} (1+e^{-z})^{-1}$

$$= -1(1+e^{-z})^{-2} \cdot \frac{d}{dz} (1+e^{-z})$$

$$= -(1+e^{-z})^{-2} \cdot -e^{-z}$$

$$= \frac{e^{-z}}{(1+e^{-z})(1+e^{-z})}$$

this is where I had to look it up online...

$$\rightarrow = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z} + 1 - 1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-z}} \cdot \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} \right)$$

$$= \frac{1}{1+e^{-z}} \cdot \left(1 - \frac{1}{1+e^{-z}} \right)$$

$$\frac{d\hat{y}}{dz} = \hat{y} \cdot (1 - \hat{y})$$

$z = w \cdot x \quad \frac{dz}{dw} = 1 \cdot x = x$

Chain Rule $\frac{dL}{dw} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{dw}$

one step $= \frac{dL}{dw} \cdot \alpha = \alpha \cdot (\hat{y} - y) \cdot \hat{y} \cdot (1 - \hat{y}) \cdot (x)$

b) Explain the role of the activation function in a neural network. Please guess without searching textbook/the Internet.

The activation function acts as a normalizer of the neuron's weighted inputs. The inputs can potentially be of great magnitude which will cause the neuron to "fire", as we want it to, but we don't necessarily want to pass the magnitude of its inputs on to the next layer. Instead of having an input with a great magnitude pass that magnitude onward to all following layers as it would if there were no activation function, instead we want to modulate the outgoing magnitude of a firing neuron by adjusting the weights of its outputs.

c) What is overfitting, and how can it be mitigated? List a couple of techniques to reduce overfitting.

Overfitting is when an algorithm essentially "memorizes" its dataset. That is a problem because when it receives a new input it has never encountered before, it will likely not be able to correctly guess the correct output. Meaning it is not able to "generalize".

This typically happens because the algorithm is overly complex, which in the case of a neural network means there's too many neurons and/or too many layers. So find ways to decrease the total number of neurons.

In the past, I've had luck by making my network "cone shaped" with larger numbers of neurons in the beginning of the layers to capture the complexity of patterns in the raw data and smaller numbers of neurons as it goes deeper into the later layers. My belief is that this shape groups the patterns it finds as it moves deeper from layer to layer.

Also, make sure to do a test / train split!

d) Can you guess what might be an underfitting problem? How can that be mitigated? List a couple of techniques to reduce underfitting.

Underfitting is when the algorithm is not complex enough, it does not have enough parameters to accommodate the complexity of the data. In neural networks, this means one should find ways to increase the number of neurons by adding more to each layer and/or adding more layers.

For example, the way neural networks that perform image recognition work is that the first layer is directly reading the pixels from the image (and the different convolutions of the image). Like for facial recognition, the first layers are attempting to group patches of skin colored pixels, then later layers are grouping different shades of skin colored patches, then more later layers are grouping the shades into the contours of facial features like a nose or a mouth or eyes, then even more later layers are grouping facial features into the correct positions to make a

face like there should be two eyes next to each other, and then ever more later still the two eyes should be positioned above a mouth, etc. There needs to be enough neurons and layers to accomplish all that.

Another way to potentially add complexity is to use multiple activation functions in each layer. Each layer could have half Sigmoid neurons and half ReLU neurons. It is something I want to experiment with, but I haven't had a chance yet. My intuition says that Sigmoid neurons shouldn't be connected ReLU neurons, so I'd like to experiment with a sort of "split brain" neural network where the layers aren't fully connected, but instead lead to two paths that aren't interconnected until the very final output layer. I'm not sure if my description is easy to follow, but it would be worthy of an experiment.

2. For the following small binary classification dataset (4 samples each with two input features (x_1, x_2) and one target feature (y)). Compute the forward passes and the gradients manually for a neural network with 1 hidden layer (with 2 neurons) and 1 output neuron.

②

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

All starting weights = 1 activation = ReLU function

Sample #1

$x_1 = 0$ Forward Pass

$x_2 = 0$

$y = 0$

$w_{11} = 1$

$w_{12} = 1$

$w_{21} = 1$

$w_{22} = 1$

$0 + 0 \rightarrow 0$

Gradients

$L = \frac{1}{2} (0 - y)^2$

$L = \frac{1}{2} (0 - 0)^2$

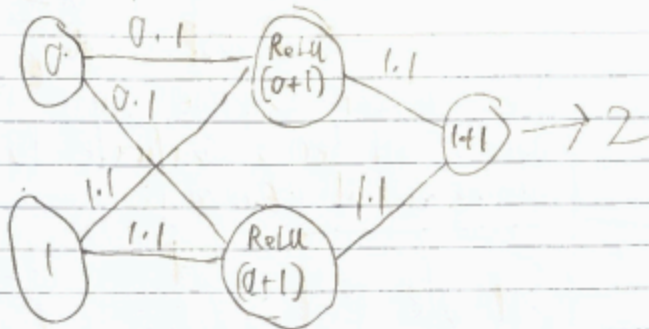
$L = 0$

Since the loss is 0, the gradient for all weights is also 0 and back propagation can be skipped for this sample.

Sample #2

Forward Pass

$$\begin{aligned}x_1 &= 0 \\x_2 &= 1 \\y &= 1 \\w_{11} &= 1 \\w_{12} &= 1 \\w_{21} &= 1 \\w_{22} &= 1 \\w_3 &= 1 \\w_4 &= 1\end{aligned}$$



Gradients

$$\begin{aligned}L &= \frac{1}{2}(\hat{y} - y)^2 \\&= \frac{1}{2}(2 - 1)^2 \\&= 0.5\end{aligned}$$

$$\hat{y} = 2$$

$$\begin{aligned}Z &= w_{21}x_1 + w_{22}x_2 \\&= 1 \cdot 1 + 1 \cdot 1 \\&= 2\end{aligned}$$

$$\begin{aligned}\frac{dL}{d\hat{y}} &= \hat{y} - y \\&= 2 - 1 \\&= 1\end{aligned}$$

$$\frac{d\hat{y}}{dz} = 1$$

$$\frac{dz}{dw_{21}} = 1 \quad \frac{dz}{dw_{22}} = 1$$

$$Z = w_{11}x_1 + w_{13}x_2$$

$$\begin{aligned}\frac{dL}{dw_{21}} &= \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{dw_{21}} \\&= 1 \cdot 1 \cdot 1 \\&= 1\end{aligned}$$

$$\frac{dL}{dw_{22}} = 1$$

$$\frac{dz}{dw_{11}} = 1 \cdot 0 + 0$$

$$\frac{dL}{dw_{11}} = 0$$

$$\frac{dL}{dw_{21}} = 1$$

$$\frac{dL}{dw_{11}} = \frac{dL}{dw_{12}} = 0$$

$$\frac{dL}{dw_{13}} = \frac{dL}{dw_{14}} = 1$$

Sample #3

almost

$$\frac{dL}{dw_{21}} = \frac{dL}{dw_{22}} = 1$$

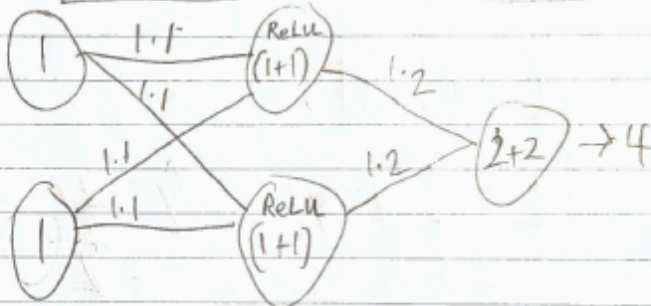
The answer is the same as Sample #2 because all terms evaluate to the same mathematical expressions.

Sample #4

$$\frac{dL}{dw_{11}} = \frac{dL}{dw_{12}} = 1$$

$$\frac{dL}{dw_{13}} = \frac{dL}{dw_{14}} = 0$$

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 1 \\ y &= 1 \end{aligned}$$



$$\begin{aligned} L &= \frac{1}{2} (\hat{y} - y)^2 \\ &= \frac{1}{2} (4 - 1)^2 \\ &= 4.5 \end{aligned}$$

$$\begin{aligned} \hat{y} &= z \\ &= 4 \end{aligned}$$

$$\begin{aligned} z &= w_{21}x_1 + w_{22}x_2 \\ &= 1.2 + 1.2 \\ &= 4 \end{aligned}$$

$$\frac{dL}{d\hat{y}} = \hat{y} - y = 3 \quad \frac{d\hat{y}}{dz} = 1 \quad \frac{dz}{dw_{21}} = \frac{dz}{dw_{22}} = 4$$

$$\frac{dL}{dw_{21}} = \frac{dL}{dw_{22}} = 3 \cdot 1 \cdot 4 = 12$$

$$\begin{aligned} z &= w_{11}x_1 + w_{13}x_2 \\ &= 1.1 + 1.1 \\ &= 2 \end{aligned} \quad \frac{dz}{dw_{11}} = \frac{dz}{dw_{13}} = 1$$

$$\frac{dL}{dw_{11}} = \frac{dL}{dw_{12}} = \frac{dL}{dw_{13}} = \frac{dL}{dw_{14}} = 3 \cdot 1 \cdot 2 = 6$$